

Project Development Phase

DATA PREPROCESSING & VISUALIZATION

Date	9 July 2026
Project Name	Food Ordering And Customer Trends
Team Member	Bhoomi Jain
Maximum Marks	4 Marks

Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the Food Ordering Behaviour dataset into Tableau Public.
- Verified that all records and fields (Order ID, Customer Age Group, Cuisine, Meal Type, City, Revenue, Customer Rating, Delivery Time, Restaurant Segment, etc.) were imported successfully.
- Confirmed that the dataset was ready for preprocessing.

Step 2: Check for Missing Values

- Examined the dataset for null or blank values.
- Verified that important fields such as Revenue, Customer Rating, Cuisine, Meal Type, and City contained valid information.
- Missing values were handled appropriately to ensure accurate analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for duplicate order records using Order ID.
- Removed duplicate entries to prevent inaccurate counts and incorrect business insights.
- Ensured that every order was represented only once.

Step 4: Validate Data Types

- Ensured that each attribute had the correct data type: Order ID (String), Revenue (Number), Customer Rating (Number Decimal), Delivery Time (Number), and Categorical attributes for Customer Age Group, Cuisine, Meal Type, and Restaurant Segment, with City as Geographic/String.

Step 5: Verify Data Consistency

- Checked for invalid or unrealistic values such as negative revenue, invalid customer ratings, or incorrect delivery times.
- Standardized cuisine names, city names, and restaurant segment values.
- Ensured consistency across all categorical fields before visualization.

Step 6: Prepare Calculated Fields

- Created calculated fields required for dashboard analysis, including: Total Orders, Total Revenue, Average Customer Rating, Average Delivery Time, Revenue by Cuisine, Revenue by City, and Order Count by Meal Type.

Step 7: Final Clean Dataset

- Verified that the cleaned dataset was complete and suitable for analysis.
- Confirmed that all preprocessing activities were completed successfully for dashboard creation in Tableau Public.

Business Questions with Visualization

S.No.	Business Question	Visualization
1	How many food orders are available for analysis after data preprocessing?	Total Order Count (KPI Card)
2	What is the total revenue generated from food orders?	Total Revenue (KPI Card)
3	Which cuisine category receives the highest number of customer orders?	Cuisine Demand Analysis
4	How does revenue vary across different cities?	City-wise Revenue Analysis
5	What is the average customer rating across restaurant segments?	Customer Rating Analysis
6	How does average delivery time vary among different restaurant segments?	Delivery Time Analysis
7	What are the ordering patterns across different meal types?	Meal Type Analysis
8	How does customer ordering behaviour differ across various age groups?	Customer Age Group Analysis

Dashboard KPIs & Delivery Visualizations

TOTAL ORDERS 50,000	TOTAL REVENUE ₹27,386,555.00	AVG ORDER VALUE ₹547.73
Delivery Time Distribution		